From a Single Axiom to Four-Dimensional Spacetime: A Readout Mechanism for Discrete Ontology and the Emergence of General Relativity

Abstract

This two-part work derives four-dimensional spacetime geometry and the Einstein field equations from a single ontological axiom: zero never stays still. Part I establishes a discrete substrate of zero-sum compensatory moves, from which we derive a dimension readout functional $F_D = \binom{D}{2} q^D$ that peaks uniquely at $D = 4$ when the participation index $q \in (1/2, 3/5)$, satisfied at $n = 16$ ($q = 0.5578$). Part II invokes the Lovelock uniqueness theorem to show that locality, second-order dynamics, and conserved source together force the Einstein field equations in four dimensions, with the spatial metric recovered via discrete Dirichlet energy and the signature $(1, 3)$ from reversible/irreversible splitting. The dimension $D = 4$ is locked by two independent routes: the graviton degree-of-freedom count $\frac{D(D-3)}{2} = 2$, and the closure–motion correspondence $\dim \ker B = \dim \operatorname{im} B$ which uniquely selects K_4 as the local fiber. All conditional inputs and open items are explicitly registered.

Contents

Part I

Part I: Dimension Readout from a Single Axiom

1 Introduction

The spacetime dimension $D = 4$ is an input rather than a conclusion within the standard physical framework [1,2]. The axiomatic systems of quantum mechanics and general relativity likewise lack a common derivation basis. Our approach is to start from a discrete substrate containing no geometry, no probability, and no action, such that dimension becomes a computable readout quantity rather than an intrinsic property of microscopic degrees of freedom.

The core stance is: dimension in discrete systems is inherently a readout quantity at the coarse-graining scale, not a microscopic degree of freedom. The framework therefore does not posit a "microscopic dimension"; it only posits a "readout dimension." The method proceeds in four steps: axiom → combinatorial structure → partition → readout.

The paper is organized as follows: §2 states the single axiom and its mathematical formulation; §3 derives words, graphs, the divergence dictionary, and zero-sum conservation; §4 establishes layered indices; §5 gives the object dictionary; §6 registers the parameter account; §7 presents the main derivation chain from balanced words to dimension readout; §8 summarizes proven results; §9 lists conditions and open items; §10 is discussion.

---

2 The Axiom and Its Mathematical Formulation

2.1 The Single Axiom

Axiom 1 (Zero Never Stays Still). The equilibrium (closed) state is not stationary; it moves continuously. Motion does not terminate at any point in time. Motion carries no weight—there is no selection rule in the system.

2.2 Mathematical Formulation

The three postulates of the axiom correspond to the following structures:

$$Z_0 = \underbrace{(\{\pm 1\},\, \iota)}_{\text{sign structure}} + \underbrace{(\mathcal{W}_n,\, \mu)}_{\text{closure constraint + counting measure}} . \tag{2.1}$$

Sign structure. Motion has two directions, giving $\{\pm 1\}$. Zero is not a stationary point but an equilibrium relation, and is therefore expressed as the quotient by an involution $\iota : \{\pm 1\} \to \{\pm 1\}$ ($\iota^2 = \mathrm{id}$, $\iota(x) \neq x$):

$$\text{zero} = \{\pm 1\}/\iota. \tag{2.2}$$

Closure constraint. Motion along a graph $\Gamma = (C, E)$ gives a divergence $d_w(v) = \#\{\text{entering } v\} - \#\{\text{leaving } v\}$ satisfying the zero-sum identity

$$\sum_{v \in C} d_w(v) \equiv 0, \qquad \text{closed} \iff d_w = 0. \tag{2.3}$$

Counting measure. No weight data exist in the system, so no motion can be singled out—every allowed motion is instantiated (full branching), with multiplicity given by counting. The only available measure on $\mathcal{W}_n$ is the uniform counting measure

$$\mu(A) = \frac{|A|}{|\mathcal{W}_n|}, \qquad |\mathcal{W}_n| = \binom{n}{n/2}. \tag{2.4}$$

Proposition 2.1 (Uniqueness of the measure). The uniform counting measure (2.4) is not an assumption.

Proof. Let μ' be any measure on $\mathcal{W}_n$. If $\mu' \neq \mu$, then there exist A, B such that $\mu'(A)/|A| \neq \mu'(B)/|B|$, i.e., there is a bias toward some motions. By the third postulate of the axiom (no selection rule), μ' is incompatible with the axiom. ■

Remark. The third postulate is a minimality argument: it establishes that no weight data exist within a single-axiom system, rather than asserting that probability is physically impossible. If probability were allowed, the full-branching conclusion would fail.

---

3 Derived Structure

3.1 Words, Graphs, and the Divergence Dictionary

Motion = one move along a directed edge of the graph $\Gamma = (C, E)$; the sequence of steps forms a word $w = (e_1, \ldots, e_n)$.

Lemma 3.1 (Update rule). Upon appending an edge $u \to v$, the divergence updates as $d \mapsto d + e_v - e_u$.

Lemma 3.2 (Zero-sum identity). For any word w, $\sum_{v \in C} d_w(v) \equiv 0$.

Proof. Each step simultaneously "enters one place and leaves another," contributing $+1 - 1 = 0$; induction on the number of steps gives the result. ■

Lemma 3.3 (Reachable set). All reachable divergences form the zero-sum hyperplane

$$\{d_w\} = \Big\{ x \in \mathbb{Z}^C : \sum_v x_v = 0 \Big\} = \operatorname{im} B =: H_Q, \qquad \dim H_Q = |C| - 1, \tag{3.1}$$

where B is the incidence matrix of the graph.

Proof. By Lemma 3.2, $\{d_w\} \subseteq H_Q$. Conversely, when Γ is connected, the image of the incidence matrix B spans H_Q, and B is totally unimodular, giving integer reachability [3]. ■

Remark. Lemmas 3.1–3.3 reduce the zero-sum constraint from an axiom to a theorem.

3.2 Full Branching and Integer Multiplicity

By the no-selection-rule postulate: no weights, probabilities, or choices exist ⇒ every allowed motion is instantiated. Multiplicity is given by counting, hence is integer. This is the watershed between our framework and "borrowed-route" approaches: the latter list full-branching generative semantics as an input; under our axiom it is a corollary.

3.3 Terminal States and Reseeding

Proposition 3.4 (Necessity of reseeding). If the active layer cannot remain empty forever (second postulate of the axiom), then finite-lifetime models must be accompanied by reseeding.

Proof. A four-model comparison shows: finite-lifetime models without reseeding necessarily extinguish; periodic-clearance plus reseeding models persist. ■

Proposition 3.5 (Geometric necessity of terminal states). Requiring a stable spectral dimension forces loop-length truncation; branches that do not close enter a terminal state at their lifetime.

Proof. In the closed-loop graph (nodes = rotation classes of zero-sum cyclic words of period T), without truncation the average degree grows unbounded with T (3.60 at $T = 8$, 9.50 at $T = 18$), leading to non-manifold discretization and no stable spectral dimension. ■

Remark. This is the only conditional derivation in the framework: if the geometric requirement is abandoned, the terminal postulate loses its justification.

---

4 Layered Indices

The physical world resides in the evolution layer. The same quantity manifests differently at different layers; layers must be aligned before comparison.

Definition 4.1 (Four-layer index).

- L0: axiom layer—postulates, integer multiplicities, local compensatory moves;
- L1: exact words (historical layer)—only increasing, irreversible;
- L1′: global closed-class layer—recording and readout;
- L2: evolution layer (active layer)—has its own update law.

Proposition 4.2 (Stratification Precedence). When two conclusions conflict, the default explanation is not contradiction but different layers; only opposites at the same layer constitute a genuine contradiction.

Reasoning. The layer index is part of the assertion; an assertion omitting the layer index is incomplete. ■

Corollary 4.3. The "readout" (π) does not occupy a layer index—it is a named input (§6).

Reasoning. If inputs were treated as layers, any conflict between two conclusions could be dissolved by "different layers," which is the opposite of the self-consistency requirement.

---

5 Object Dictionary

5.1 Kinematic Objects

Symbol	Name	Definition
$\Gamma = (C, E)$	graph	positions and steps; finite, connected
$w = (w_1, \dots, w_n)$	word	sequence of steps
d_w	divergence	entering count − leaving count
H_Q	zero-sum hyperplane	$\{x : \sum x_v = 0\}$
$\mathcal{W}_n$	balanced word space	$\{w \in \{\pm 1\}^n : \sum w_k = 0\}$
s_k	prefix sum	$\sum_{i \le k} w_i$, $s_0 = s_n = 0$

5.2 Intrinsic Quantities

$$\text{height} : h(w) := \max_{0 \le k \le n} |s_k|, \qquad \text{area} : a(w) := \sum_{k=1}^{n} s_k. \tag{5.1}$$

Proposition 5.1. Both h and a are anchored at $s_0 = 0$ (i.e., zero), not at n.

Proof. Both are given directly by the prefix-sum sequence, which starts at $s_0 = 0$. ■

5.3 Readout Objects

Symbol	Name	Definition
π	partition (readout)	$\pi : \mathcal{W}_n \to \{1, \dots, k\}$
$\mathcal{B}_b$	block	$\{w : \pi(w) = b\}$
ω_b	block weight	$\lvert\mathcal{B}_b\rvert / \lvert\mathcal{W}_n\rvert$
q	participation index	$\sum_b \omega_b^2$
F_D	dimension-selection functional	$\binom{D}{2} q^D$
λ	normalized spectrum	top three block weights, descending and normalized

---

6 Parameter Account

$$Z_0 + \underbrace{\underset{\sim}{n}}_{\text{free parameter}} + \underbrace{\underset{\sim}{\pi}}_{\text{structural input}} + \underbrace{\underset{\sim}{\kappa}}_{\text{scale input}} = \text{complete framework.} \tag{6.1}$$

That is: one axiom, one free parameter, two named inputs.

Item	Type	What it buys
Z_0 (three postulates)	axiom	words/graph/steps; closure; full branching; integer multiplicities
n	free parameter	terminal scale; entry point for blocking
π	named input	dimension-selection peak; contextuality threshold; module flow type
κ	named input	scale (1 step = how many physical units)
$\beta\varepsilon$	parameter	effective ladder temperature (height cost), affects state layer only
Γ	named input	topological carrier; specialized to path graph $\mathbb{Z}$ in this work

Proven non-derivable items: absolute scale, D itself (all metric-side structures are graph-independent, so low dimensions cannot be excluded), measure normalization, absolute amplitude.

---

7 Main Derivation Chain

7.1 Kinematic Starting Point

Specializing to the path graph:

$$\Gamma = \text{path graph } 0, \pm1, \pm2, \dots; \quad \text{step} = \pm1; \quad \text{closed} \iff s_n = 0. \tag{7.1}$$

$$\mathcal{W}_n = \Big\{ w \in \{\pm1\}^n : \sum_{k=1}^{n} w_k = 0 \Big\}, \qquad |\mathcal{W}_n| = \binom{n}{n/2}. \tag{7.2}$$

Remark. $\mathcal{W}_n$ is a set selected by the zero-sum constraint—no time, no probability, no space.

7.2 Partition and Self-Identification Criteria

Partition $\pi : \mathcal{W}_n \to \{1, \dots, k\}$, block weights $\omega_b = |\mathcal{B}_b|/|\mathcal{W}_n|$.

Definition 7.1 (Self-identification criterion). Partition labels must be recoverable from the block itself (after closure the system has no memory; labels cannot be carried by memory). This requires three structural criteria:

1. **Locally Computable**: the primitive $h(w)$ can be computed from a single word;
2. **Anchored at zero**: $h(w)$ is referenced to $s_0 = 0$, not to n;
3. **Non-Vanishing Bulk**: $\max_b \omega_b$ does not decay with n.

Proposition 7.2. The family of candidate partitions satisfying these three criteria includes but is not limited to: height-level partition (hmax), longest-run partition (maxrun), first-flip partition (firstflip), and flip-count partition (nflip).

Proof. Direct verification from the criteria. ■

Remark. Criteria 1 and 2 constrain the single-word primitive $h(w)$; the merging threshold $t^* = \arg\max_h b_h$ (Eq. 7.6) is an ensemble-level coarse-graining rule that depends on the global mode and n, not on individual words. The two are at different layers and do not conflict.

7.3 Height Blocks and the Reflection Principle

$$b_h = \big|\{w \in \mathcal{W}_n : h(w) = h\}\big| = N_{\leq}(n,h) - N_{\leq}(n,h-1). \tag{7.3}$$

The reflection principle under two-sided absorbing boundaries $\pm(h+1)$ gives [4]:

$$N_{\leq}(n,h) = \sum_{m\in\mathbb{Z}} (-1)^m \binom{n}{n/2 + m(h+1)}. \tag{7.4}$$

Explicit computation for $n = 16$:

		h	1	2	3	4	5	6	7	8		
b_h	256		4118		4858		2518		880		208	30	2

$$\sum_h b_h = 12870 = \binom{16}{8} \quad \checkmark \tag{7.5}$$

Merging rule (taking the modal level as threshold):

$$t^* := \arg\max_h b_h; \quad h > t^* \text{ form independent blocks;} \quad h \leq t^* \text{ merge into one block.} \tag{7.6}$$

For $n = 16$, $t^* = 3$, and the unnormalized block sizes are

$$[|\mathcal{B}_b|] = [9232,\ 2518,\ 880,\ 208,\ 30,\ 2], \qquad k = 6. \tag{7.7}$$

7.4 Three Quantum Gates

Let $\lambda = (\lambda_1, \lambda_2, \lambda_3)$ be the top three block weights in descending order, normalized, and $q = \sum_b \omega_b^2$.

Definition 7.3 (Three gates).

- **Gate 1 (survival window)**: $q \in (1/2,\ 3/5)$;
- **Gate 2 (contextuality)**: $S_{\max} = \mu_1\lambda_1 + \mu_2(\lambda_2 + \lambda_3) > 2$, where $\mu_1 = \sqrt{5}$, $\mu_2 = (5-\sqrt{5})/2$;
- **Gate 3 (density)**: rank{prime exponent difference vectors} $= k - 1$.

The μ coefficients in Gate 2 are the von Neumann trace inequality coefficients of the KCBS inequality [5,6]; the non-contextual bound is 2.

7.5 Dimension-Selection Functional and Peak Condition

$$F_D = \binom{D}{2} q^D. \tag{7.8}$$

Here $\binom{D}{2} = \dim \mathfrak{so}(D-1,1)$ (the dimension of the Lorentz algebra, i.e. the number of independent Lorentz generators), and q^D is the cost of requiring simultaneous record across D directions.

Lemma 7.4 (Peak condition).

$$\frac{F_D}{F_{D-1}} = \frac{D}{D-2} q \quad \Longrightarrow \quad \text{unique finite maximum at } D = 4 \iff q \in (1/2,\, 3/5). \tag{7.9}$$

Proof. $F_D/F_{D-1} > 1 \iff q > (D-2)/D$. At $D = 4$ this requires $q > 1/2$; at $D = 5$, $q < 3/5$. Thus $q \in (1/2, 3/5)$ simultaneously gives $F_4 > F_3$ and $F_5 < F_4$; for $D \geq 5$, $(D-2)/D \geq 3/5 > q$, so F is strictly decreasing for $D \geq 5$. At the boundary: $F_0 = F_1 = 0$, $F_2 = q^2$, $F_3 = 3q^3$, and when $q > 1/2$, $F_3/F_2 = 3q > 3/2 > 1$, so $D \leq 3$ does not exceed F_4. The maximum is unique. ■

7.6 Layer Distinction

The three gates read different layers: the blocking layer (k, ω_b, q) is determined by pure counting and does not depend on $\beta\varepsilon$; the state layer (spectrum of ρ, $S_{\max}$) is determined by the measure $\omega \propto m\, e^{-\beta\varepsilon t}$ and varies monotonically with $\beta\varepsilon$.

Consistency check. At $n = 16$, $k = 6$ for all $\beta\varepsilon$; $S_{\max}$ increases monotonically from 2.006279 ($\beta\varepsilon = 0$) to 2.235986 ($\beta\varepsilon = 2$).

7.7 7.7 Explicit Solution for $n = 16$

$$q = \frac{9232^2 + 2518^2 + 880^2 + 208^2 + 30^2 + 2^2}{12870^2} = \mathbf{0.55777859} \in (1/2,\, 3/5) \quad \checkmark \tag{7.10}$$

$$(\lambda_1, \lambda_2, \lambda_3) = (0.73095804,\ 0.19936659,\ 0.06967538), \tag{7.11}$$

$$S_{\max} = \sqrt{5}\,(0.73095804) + \frac{5-\sqrt{5}}{2}(0.19936659 + 0.06967538) = \mathbf{2.00627871} > 2 \quad \checkmark \tag{7.12}$$

$$\text{rank} = 5 = k - 1 \quad \checkmark \tag{7.13}$$

7.8 Full-Chain Census

$$Z_0 \longrightarrow \mathcal{W}_n \longrightarrow \pi \longrightarrow (n, \pi) \longrightarrow D = 4. \tag{7.14}$$

A census over $n = 6, \ldots, 48$ (22 even values) $\times$ 4 self-identification candidate partitions (88 combinations) yields:

$$(n, \pi) \in \{(16, \texttt{hmax}),\ (20, \texttt{maxrun})\} \qquad (2/88 = 2.3\%). \tag{7.15}$$

---

8 Summary of Proven Results

1. **One axiom**, three postulates each buying one thing (proven).
2. **Zero-sum constraint is a theorem, not an axiom** (Lemmas 3.1–3.3, proven).
3. **Counting measure is unique** (Proposition 2.1, proven).
4. **Reseeding is forced by the requirement of persistent motion** (Proposition 3.4, proven).
5. **Terminal states are forced by geometric stability** (Proposition 3.5, conditionally derived).
6. **Self-identification criteria restrict the candidate partition family** (Proposition 7.2, proven).
7. **The height-level partition passes all three quantum gates and peaks at $D = 4$** (§7.7, computed).
8. **Peak at $D = 4 \iff q \in (1/2, 3/5)$** (Lemma 7.4, proven).
9. **D cannot be determined from proven structures; absolute scale is non-derivable** (§6, proven non-derivable).
10. **Second route**: the cycle space $Z_1(K_4)$ and displacement space H_Q form an axial–polar vector duality in three dimensions via the Hodge star $\star : \Lambda^2\mathbb{R}^3 \to \mathbb{R}^3$, independently locking $d = 3$ without the F_D peak condition (see companion paper [7]).

---

9 Conditions and Open Items

This section is of equal importance to the main text.

#	Condition	Status
1	**Quantumness required** (Gate 2: $S_{\max} > 2$)	Premise. Critical bottleneck of the entire chain. Necessity unproven
2	**Partition criterion non-unique**	Proven non-unique: the three structural criteria are simultaneously satisfied by hmax and firstflip
3	Necessity of the three structural criteria	Unproven (but all are preference-free)
4	Exhaustiveness of the candidate partition family	Unproven (only 4 self-identification candidates checked)
5	**Conditionality of $D = 4$**	Not true for all n: only $n \in \{16, 20\}$ give peak $= 4$. The Hodge duality route independently locks $d = 3$, but requires the three-dimensional geometric realization premise
6	Scan range for n	Only $n = 6, \ldots, 48$

#	Condition	Status
7	Status of κ	Fixed once or reselected each round; indistinguishable within the framework
8	Arrow of time	Still open
9	Continuum limit / metric / field equations	Not addressed in this work; see companion paper [7]
10	**Truncation renormalization in Gate 2**	$S_{\max} > 2$ is obtained by taking the top three blocks and renormalizing (discarding $h = 6, 7, 8$, total 240 words). Without renormalization $S_{\max} = 1.9946 < 2$, failing the contextuality bound. Gate 2 is currently conditionally justified
11	**Gate 2 fixed at** $\beta\varepsilon = 0$	$\beta\varepsilon$ is a free parameter, but the census uniqueness holds at $\beta\varepsilon = 0$ (pure counting); otherwise any partition passing Gates 1 and 3 could have $S_{\max}$ cross 2 by taking $\beta\varepsilon > 0$
12	**Functional form** $F_D = \binom{D}{2} q^D$	Taking the number of Lorentz generators times q^D and choosing the physical dimension by the maximum is a readout functional hypothesis on par with π, not a theorem derived from Z_0

10 Discussion

$$Z_0 \Longrightarrow \mathcal{W}_n \Longrightarrow \pi \text{ (self-identifying)} \Longrightarrow D = 4.$$

All results are conditional on quantumness; the axiom itself is unchanged.

The framework's free content consists of exactly four items: one axiom Z_0, one free parameter n, two named inputs π and κ.

The two weakest links at present are: link (2) (partition non-uniqueness, proven) and the quantumness premise of link (3) (unproven). Subsequent work targets these two links, as well as establishing the continuum limit and field-theoretic stage.

Claims not made: Z_0 has been modified; the chain holds unconditionally; π has been uniquely determined; $D = 4$ holds for all n; contextuality resolves the arrow of time; this work provides a

continuum limit, metric, or gravitational field equations.

References

[1] R. Penrose, *The Road to Reality: A Complete Guide to the Laws of the Universe*, Vintage Books, 2005.

[2] C. J. Isham, "Structural issues in quantum gravity," in *Recent Perspectives in the Quantum Theory of Gravity*, arXiv:gr-qc/9510034, 1994.

[3] A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986.

[4] W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968 (reflection principle and first-passage distributions).

[5] A. A. Klyachko, M. A. Can, S. Binicioğlu, A. S. Shumovsky, "A simple test for hidden variables in spin-1 systems," *Phys. Rev. Lett.* **101**, 020403 (2008).

[6] R. W. Spekkens, "Evidence for the epistemic view of quantum states: A toy theory," *Phys. Rev. A* **75**, 032110 (2007).

[7] Companion paper: *Conditional Reduction to General Relativity from Zero-Sum Constraints: Lovelock Uniqueness and the Group-Theoretic Origin of Dimension* (in preparation).

Part II

Part II: Conditional Reduction to General Relativity

11 Introduction

The spacetime dimension $D = 4$ is an input rather than a conclusion within the standard physical framework [1,2]. Our approach starts from a discrete substrate containing no manifold, coordinates, metric, probability, or action, such that the shape of the Einstein field equations becomes a consequence derivable from the Lovelock uniqueness theorem.

Strategy. We do not assemble GR piece by piece. GR is compressed into three structural properties: (L) locality, (O) second-order, (C') conserved source. The substrate need only provide these three; the shape of the geometric field equations is then locked by the Lovelock theorem [3].

Scope. The Lovelock theorem's literal statement is "in 4 dimensions." This lever therefore cannot be used to select $D = 4$; dimension must be argued separately (§6).

---

12 Core Postulates of the Substrate

The substrate declares only the following; no manifold, coordinates, dimension, distance, curvature, action, or probability weights appear:

- **State**: codimension-1 affine constraint on integer lattice (zero-sum);
- **Transport**: connected, reversible compensatory moves;
- **Energy**: Dirichlet-type quadratic form;
- **Time**: step index τ and terminal;
- **Record**: multiplicity multiset of closed classes;
- **Weights**: integer multiplicities.

The substrate axiom is $Z_0 = (\{\pm 1\}, \iota) + (\mathcal{W}_n, \mu)$, where ι is the direction involution and μ the uniform counting measure [4].

---

13 Algebraic Skeleton

13.1 Zero-Sum Codimension and Transport Rank

Lemma 3.1 (Codimension 1). The zero-sum constraint $\sum_v x_v = 0$ gives only one linear condition:

$$\dim H_Q = m - 1, \qquad \{e_j - e_1\}_{j=2}^{m} \text{ is a basis.}$$

Corollary 3.2. To obtain "four dimensions," one must first have $m - 1 = 4$ or $m = 4$; neither is a consequence of the substrate postulates.

Lemma 3.3 (Transport rank and integer reachability).

$$\operatorname{im} B = H_Q, \qquad \operatorname{rank} B = m - 1,$$

$$x \in H_Q \cap \mathbb{Z}^C \implies \exists f \in \mathbb{Z}^E,\ Bf = x.$$

Proof. B is totally unimodular [5]; integer reachability follows from standard integer linear programming results. ■

13.2 Discrete Dirichlet Energy

Lemma 3.4. The discrete Dirichlet energy with edge weights K_e

$$E(f) = \sum_e K_e (f_i - f_j)^2 \geq 0, \qquad E(f) = 0 \iff f \text{ is constant},$$

i.e. $E = f^{\mathsf{T}} L f$, $\ker L = \operatorname{span}\{\mathbf{1}\}$, $L(H_Q) \subseteq H_Q$, $L|_{H_Q} \succ 0$.

Lemma 3.5 (No preference ⇒ uniform conductance). If Γ is edge-transitive, then $K_e \equiv \kappa$.

Remark. This is conditional: path graphs P_4, P_5 have 2 edge orbits, and no-preference does not yield uniform conductance.

13.3 Local Quadratic Form and Spatial Metric

Lemma 3.6 (Metric-volume consistency inversion). Continuum approximation gives $E \to \int Q^{ij} \partial_i f\, \partial_j f\, \omega$. Requiring metric-volume consistency

$$\mu = \operatorname{vol}_h = \sqrt{\det h}\, \omega, \tag{3.1}$$

one obtains $Q = \sqrt{\det h}\, h^{-1}$, $\det Q = (\det h)^{d/2-1}$ (d = spatial dimension). For $d \neq 2$ the unique inverse is

$$h = (\det Q)^{\frac{1}{d-2}}\, Q^{-1}. \tag{3.2}$$

For $d = 2$, $\det Q = 1$, and $h \mapsto \Omega^2 h$ leaves Q invariant; only the conformal class remains.

Remark. (3.1) is the sole additional input in this step; it binds the Dirichlet quadratic form one-to-one to the spatial metric and removes conformal ambiguity for $d \neq 2$. The unique inverse holds for all $d \neq 2$ and does not select any dimension.

13.4 Splitting, Time, and Signature

Lemma 3.7 (Reversible/irreversible splitting).

$$T \cong H_Q \oplus \mathbb{R}_\tau \cong \mathbb{R}^m.$$

The regular bilinear form can only be $g = d\tau^2 - h$ or $g = -d\tau^2 + h$. Taking the former gives signature$(g) = (1, m-1)$.

Mechanism. The substrate's compensatory moves are reversible (each edge has an inverse move); step progression is irreversible. Spatial directions carry the inverse structure; the step direction does not. The signature originates here—there is no need to single out a time direction among the spatial directions.

Boundary. $1 \oplus (m-1)$ and $h > 0$ are derived; which side carries the minus sign is a sign convention.

13.5 Curvature and Two-Derivative Uniqueness

Lemma 3.8. Given a smooth manifold M and nondegenerate g, the connection $\Gamma^a{}_{bc}$, curvature $R^a{}_{bcd}$, Ricci R_{ab}, and Einstein tensor G_{ab} are uniquely determined by standard differential geometry [6].

Lemma 3.9 (Uniqueness within the two-derivative family). Let $E_{ab} = \alpha R_{ab} + \beta R g_{ab} + \gamma g_{ab}$. Using $\nabla^a g_{ab} = 0$ and the contracted Bianchi identity $\nabla^a R_{ab} = \frac{1}{2}\nabla_b R$:

$$\nabla^a E_{ab} = \Big(\frac{\alpha}{2} + \beta\Big)\nabla_b R. \tag{3.3}$$

Thus when $\nabla R \not\equiv 0$, divergence-free $\iff$ $\beta = -\alpha/2$, i.e. $E_{ab} \in \mathrm{span}\{g_{ab}, G_{ab}\}$.

13.6 Lovelock Uniqueness and the GR Criterion

Lemma 3.10 (Lovelock uniqueness) [3]. In 4 dimensions, a symmetric second-rank tensor constructed from g and its derivatives up to second order, linear in the highest derivative, and satisfying $\nabla^a E_{ab} = 0$ must be

$$E_{ab} = c_1 G_{ab} + c_2 g_{ab}.$$

(The Gauss–Bonnet term is a topological term in 4 dimensions and does not enter the field equations.)

Corollary 3.11 (GR criterion).

$$(L) \wedge (O) \wedge (C') \implies G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab}. \tag{3.4}$$

Here (C') requires that T_{ab} be a functional of matter data (the metric only enters index raising/lowering and normalization) and must not contain independent metric degrees of freedom.

Remark. The Lovelock theorem's literal premise "in 4 dimensions" means it cannot be used to derive $D = 4$; dimension must be argued separately (§6).

13.7 Locality Bifurcation and Excluded Routes

Lemma 3.12. The discrete Dirichlet energy is local; the effective resistance distance $R_{ij} = (e_i - e_j)^\mathsf{T} L^+ (e_i - e_j)$ is nonlocal.

Lemma 3.13 (Limitations of the algebraic assembly route). The curvature of an equilateral complex is locked by combinatorial structure: the deficit angles around edge k are $+148.41°, +77.88°, +7.36°, -63.17°$ ($k = 3, 4, 5, 6$), with only $k = 3, 4, 5$ giving positive deficit. Algebraic assembly can therefore only land on an equilateral, rigid subset and cannot approximate a general metric.

Excluded route	Basis
Algebraic assembly ($K \to h$)	Unique fixed point = uniform ⇒ equilateralization ⇒ one-parameter rigidity
Effective resistance as geometry	Nonlocal; external edges break local isotropy
Homogeneity degree for dimension selection	Homogeneity degree is 1 for every d; does not select dimension

---

14 Metric Construction and the Lovelock Lever

14.1 Common Weight Formula

All three metric routes use the same weight formula

$$w_{ij} = \phi_i \phi_j A_{ij}, \qquad h = \text{graph Laplacian with weight } w \text{ restricted to } H_Q, \tag{4.1}$$

differing only in the source of ϕ.

Route 1 (closed-walk counting, $k \to \infty$). The total traversal count of edge (i, j) by length-k closed walks is

$$N_{ij}^{(k)} = k\, A_{ij} (A^{k-1})_{ij}. \tag{4.2}$$

After normalization, $w_{ij} \propto \phi_i \phi_j A_{ij}$, with ϕ the Perron vector. Zero free parameters. But w depends on the Perron eigenvector—a global quantity of the entire graph—so (L') (g is locally determined by data) fails.

Route 2 (finite truncation $k = n$). Truncating (4.2) at scale k:

$$w_{ij}^{(k)} = \sum_{m=2}^{k} m\, A_{ij} (A^{m-1})_{ij}. \tag{4.3}$$

Locality holds exactly when $k \leq$ graph diameter (the effect of distal perturbations on central edge weight ratios is exactly zero). $k > n$ counts unrealizable closed words; $k < n$ misses realizable ones, so $k = n$ is forced.

Route 3 (locally given ϕ). If ϕ is given locally, (L') holds exactly, and all three premises are satisfied.

Remark (Scope of the (O) and (C) conditions). The second-order (O) and conservation (C) checks in Routes 1–3 strictly verify that the graph Laplacian L is a second-order, charge-conserving operator on scalar functions ($\ker L = \mathrm{span}\{\mathbf{1}\}$), which guarantees that the emergent spatial metric h is well-defined. They do not automatically imply that the field tensor $E_{ab}[g]$ is divergence-free under full four-dimensional spacetime diffeomorphisms; four-dimensional covariance $\nabla^a E_{ab} = 0$ remains an essential macroscopic symmetry postulate registered in §7.1 (Item 8).

14.2 Degrees of Freedom and Parameterization

The number of spatial metric components is

$$\dim \mathrm{Sym}^2 H_Q = \binom{m}{2} = \frac{(m-1)m}{2}. \tag{4.4}$$

Key distinction. (4.4) counts spatial metric h components ($m = 4$, 3D space: 6). Our construction $g = d\tau^2 - h$ (ADM $N = 1, \beta = 0$) fixes $g_{00} = 1$, $g_{0i} = 0$; graph edge weights provide only the 6 components of h, never the 10 components of the 4D spacetime metric g_{ab}.

Parameterization is not bijective. The weight formula (4.1) uses a vertex field $\phi = (\phi_1, \ldots, \phi_m)$ to give edge weights $w_{ij} = \phi_i \phi_j A_{ij}$. ϕ has only m free parameters, while $\dim \mathrm{Sym}^2 H_Q = \binom{m}{2}$; on K_m, the $\binom{m}{2}$ edge weights from m vertex values always satisfy the cross-product constraint $w_{12}w_{34} = w_{13}w_{24} = w_{14}w_{23} = \phi_1\phi_2\phi_3\phi_4$. It therefore spans only a $\leq m$-dimensional submanifold of $\mathrm{Sym}^2 H_Q$, not a bijection. The injectivity of $w \mapsto h$ (for independent weights) holds, but the ϕ parameterization of (4.1) is a local parameterization on a submanifold, not proven to cover the full space.

Graviton degrees of freedom. In general 4D GR: 10 metric components -4 coordinate gauge -4 initial-value constraints $= 2$ physical propagating degrees of freedom, i.e. $\frac{D(D-3)}{2} = 2$.

14.3 Refinement Convergence and Nonlocality of the Limit

The edge-weight ratio converges with truncation k, but extremely slowly, and the limit is the Perron form $P_1 = vv^{\mathsf{T}}$, globally nonlocal. The deviation decay length is

$$\tau \approx \frac{2N^2}{3\pi^2}. \tag{4.5}$$

Conclusion. The effective metric has a limit under refinement; the limit is the Perron form, nonlocal $\Rightarrow$ there exists no UV-fixed local metric.

---

15 Continuum Limit and Causal Structure

15.1 5.1 Γ-Convergence

Lovelock requires Γ-convergence (local structure convergence), not a UV fixed point (self-similarity) [7]. A standard fact of homogenization theory: non-self-similar coefficient families still Γ-converge.

With fixed step count $k = 8$, the profile maximum difference between adjacent N converges at approximately $O(a^2)$.

Boundary. Γ-convergence is not proven (no compactness, no Γ-liminf); only convergence and order are measured, and only on a 1D chain with a single profile.

15.2 Causal Structure: Support Cone and Cone Edge

Finite causal velocity requires saturation (per-point capacity), not current relaxation. Out-of-cone signals are exponentially small but nonzero pulled fronts [8].

The support cone is always exact: nearest-neighbor transport expands at most one lattice site per step, $\operatorname{supp}\rho_n \subseteq [-n, n]$, independent of B.

Cone-edge visibility is derived analytically: on the $e^{-\mu x}$ mode, the one-step mapping characteristic equation is

$$\Lambda^2 = B\cosh^2\mu \implies \lambda(\mu) = \tfrac{1}{2}\log B + \log\cosh\mu. \tag{5.1}$$

The cone-edge density decay rate is

$$\varepsilon(B) = \tfrac{1}{2}\log\Big(\tfrac{4}{B}\Big). \tag{5.2}$$

Visible cone edge $\iff B = 4$. $B = 4$ is upgraded from a model parameter to a self-consistency condition.

---

16 Dimension Account

This is the weakest link of the entire chain.

16.1 The Core Postulates Do Not Constrain Dimension (No-Go)

Proposition 6.1 (Classification-level no-go). Let $\mathcal{A}$ be the substrate core postulates. Then for every integer $m \geq 2$, there exists a model $\mathcal{M}_m$ of $\mathcal{A}$ with channel-set cardinality $|C| = m$.

Proof. Take $C = \{0, \ldots, m-1\}$, $E = \{\{i, i+1\} : i \in \mathbb{Z}_m\}$ (cycle graph C_m). Verify postulate by postulate: C_m is finite and connected; every move has an inverse; all $2m$ directed moves are taken with equal weight; step index progresses irreversibly; closed state $x = 0$ exists. Numerical verification ($m = 2, \ldots, 12$): $\operatorname{rank} B = m - 1$, $\ker L = \operatorname{span}\{\mathbf{1}\}$, $L|_{H_Q} \succ 0$. C_m is edge-transitive, so the no-preference ⇒ uniform conductance premise also holds. ■

$$\implies \text{any statement depending only on } D \text{ cannot be derived from the substrate core postulates.} \tag{6.1}$$

16.2 Excluding Low Dimensions

D	Conclusion	Basis
2	No gravitational dynamics	$R_{ab} = \frac{1}{2} R g_{ab} \implies G_{ab} \equiv 0$
3	No graviton	Weyl tensor vanishes identically; curvature fully determined by local matter
4	Propagating graviton exists	see §6.3

16.3 6.3 Uniqueness of $D = 4$ from Graviton Degrees of Freedom

The D-dimensional graviton degrees of freedom are $\frac{D(D-3)}{2}$:

D	2	3	4	5	6	7	
$\frac{D(D-3)}{2}$		-1	0	**2**	5	9	14

$$\frac{D(D-3)}{2} = 2 \iff D^2 - 3D - 4 = 0 \iff D \in \{-1, 4\} \implies D = 4. \tag{6.2}$$

Screening chain. $\{2, 3, \dots\} \xrightarrow{D(D-3)/2>0} \{4, 5, \dots\} \xrightarrow{\text{minimality}} 4.$

Boundary. Minimality, even-dimension reading, and sign–polarization identification are not consequences of the substrate postulates.

16.4 Candidate Constraints as Synonymous Reformulations

Of 15 candidate constraints tested item by item, items 1–10 all have survival set $\{4\}$; they are all synonymous reformulations of $(Z_2)'$ and do not constitute independent evidence.

Among them, item 7, $b_1(K_m) = m - 1$, is examined in detail below.

16.5 Representation-Theoretic Status of the Cycle–Displacement Duality

As an **oriented** cycle space, $Z_1(K_4)$ decomposes as the standard S_4-module $(2, 1, 1) \cong (3, 1) \otimes \mathrm{sgn}$, which is the representation of axial vectors (pseudovectors, flux, magnetic field). The displacement space is $H_Q = (3, 1)$, the standard polar-vector representation. The two are distinct S_4-modules (they differ by the sign representation). **However, in three-dimensional geometric realization** (K_4 embedded as a regular tetrahedron in $\mathbb{R}^3$), the Hodge star operator

$$\star : \Lambda^2 \mathbb{R}^3 \longrightarrow \mathbb{R}^3$$

establishes a canonical duality between the axial cycle space and the polar displacement space. This is not a numerical coincidence: it is the group-theoretic counterpart of §6.8(ii) "antisymmetric composition closure"—the identity $\Lambda^2 \mathbb{R}^3 \cong \mathbb{R}^3 \otimes \mathrm{sgn}$ is precisely the Hodge duality between closed 2-forms (loops, flux) and 1-forms (displacements) in three dimensions [9].

16.6 Dimension-Blind No-Go

All four metric-side structures $u = h/\sqrt{n}$, $\mathrm{H}(w)$, $\min(k, n-k)$, $T(M, n)$ do not reference the adjacency matrix A and take the same value on any graph.

$$\Longrightarrow \textbf{ the metric side cannot exclude low dimensions (e.g., } d = 1). \tag{6.3}$$

16.7 The Closure–Motion Correspondence Principle

The core postulates do not constrain dimension (Proposition 6.1). What fixes the dimension is this correspondence:

> **Closure–motion correspondence [named input].** The axiom, read at the counting level: $\dim\ker B = \dim\mathrm{im}\, B$ (degrees of freedom of closed loops = degrees of freedom of motion).

Here B is the incidence matrix of compensatory moves: $\mathrm{im}\, B = H_Q$ is the zero-sum state, and $\ker B$ is the closed-loop space.

$$\text{dictionary}: \quad d = m-1, \quad \text{uniquely fixed by } \binom{m}{2} - (m-1) = \binom{m-1}{2}; \tag{1}$$

$$\text{reading}: \quad \binom{m}{2} - (m-1) = m-1 \iff m = 4 \quad (\text{unique}); \tag{2}$$

$$\text{no preference (edge-transitive)}: \quad \deg \text{ equal} \Longrightarrow t\, m = 4(m-1) \iff (\deg, m) = (3, 4). \tag{3}$$

Machine check: on K_m, $\dim\ker B = \binom{m}{2} - (m-1)$ and $\dim\mathrm{im}\, B = m-1$ are equal only when $m = 4$ ($m = 2$ gives $0 \neq 1$, $m = 3$ gives $1 \neq 2$, $m \geq 5$ always gives $\dim\ker B > \dim\mathrm{im}\, B$).

Together the three give: **spatial dimension $d = 3$, channel count $m = 4$,** $\deg = 3$**, spacetime dimension** $D_{\mathrm{st}} = d + 1 = 4$.

$D_{\mathrm{st}} = 4$ follows from two independent routes: the substrate-side premise is the closure–motion correspondence, th (6.5)

Why the correspondence cannot be a consequence of the postulates. The substrate gives $\Gamma = (C, E)$ together with "motion"; the postulates only constrain *how* to move—"full branching" means every *allowed* motion is instantiated, and does not constrain which pairs have allowed moves. Any condition of the form "Γ must be complete/self-dual" is a choice of Γ, not a consequence of the postulates. Counterexample: cycle graph C_m ($m \geq 4$) satisfies the postulates and edge-transitivity, but $\dim\ker B = 1 \neq m-1$.

16.8 Equivalent Formulations of the Correspondence Principle

The counting content $\binom{d}{2} = d$ has three equivalent formulations:

(i) Pairing uniqueness. Each edge has exactly one opposite edge $\iff \binom{m-2}{2} = 1 \iff m = 4$.

(ii) Antisymmetric composition closure. $\dim \Lambda^2 V = \dim V \iff D = 3$: the "relative relation" of two directions is again a direction (cross-product structure). I.e. $\binom{D}{2} = D \iff D \in \{0, 3\}$, with $D = 3$ the unique nontrivial solution.

(iii) A_3 self-duality. The 6 edge directions $\pm(e_i - e_j)$ of K_4 are exactly the 6 positive roots of A_3; $\mathrm{Aut}(K_4) = S_4$ is the Weyl group of A_3. Among A_n, only A_3 satisfies "number of relations = dimension": $\binom{n+1}{2} - n = n \iff n = 3$. This is precisely the accidental isomorphism $A_3 \cong D_3$ ($\mathfrak{so}(3) \cong \mathbb{R}^3$ cross-product isomorphism).

$$\implies \Gamma = K_4 : \text{ 4 vertices, deg} = 3, \text{ 6 edges, 3 opposite-edge pairs; angle} = \arccos(-1/3) = 109.4712°; \ \mathrm{Aut}(K_4) = \tag{6.6}$$

---

17 Axiomatic Budget

17.1 What the Lovelock Lever Presupposes

#	Presupposition	Nature
1	$D = 4$ (literal hard-coding)	Cannot select dimension via this lever
2	Metric exists and is smooth	Input
3	Metric-volume consistency $\mu = \mathrm{vol}_h$	Sole additional input of (3.1)
4	Second-order (O)	Excludes higher-order terms
5	Conserved source (C')	(C) as-is is vacuous
6	Source = relativistic matter	Additional claim
7	Metric nondegenerate and Lorentzian	Input
8	4D diffeomorphism covariance $\nabla^a E_{ab} = 0$	Input: the preferred foliation of $g = d\tau^2 - h$ gives only 3D spatial diffeomorphism invariance

17.2 Absence of Intrinsic Scales

The dimensionful degrees of freedom in the geometric sector are exactly 1. (7.1)

Evidence. The layer coefficient (edge $(0, 1)$) $\binom{m-1}{m/2}$ is an integer ($m = 2, 4, 6, 8$ gives $1, 3, 10, 35$). All raw metric-side data are integers $\Rightarrow$ they cannot carry length.

Unit-transformation invariance. Under $W \to sW$ (s spanning 8 orders of magnitude), normalized eigenvalue deviation is 1.89×10^{-15}, edge-weight ratio deviation is 0.

On G, Λ. Dimensional constants are units, not parameters [10]. "Deriving the numerical values of $G, \Lambda, \hbar$" is therefore not a well-posed problem.

---

18 Main Results and Boundaries

18.1 Main Result

$$\begin{aligned} &\text{substrate (zero-sum constraint)} + \text{(inputs: embedding / source class / scale)} \\ &\quad \Longrightarrow (L) \wedge (O) \wedge (C') \\ &\quad \Longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab} \quad (\text{in } D = 4). \end{aligned} \tag{8.1}$$

the shape of the field equations is uniquely forced; the metric and dimension are inputs; the source is a no-go. (8.2)

18.2 Layering of "Dimension Is an Input"

"Input" here refers specifically to the Lovelock layer: the shape theorem $E_{ab} = \alpha G_{ab} + \beta g_{ab}$ takes D as an argument, and Lovelock itself does not select dimension. But the value $D = 4$ is not a bare free input; it is given by the readout chain: peak condition locks $L = 4$ (Gate 1) + closure-equality $\dim\ker B = \dim\mathrm{im}\, B$ locks K_4/A_3. This readout chain is half-locked: the peak condition hard-presses to $L = 4$; the step "$L = 4$ is exactly the physical spacetime dimension" is only count-level equality—$Z_1(K_4) \cong (2,1,1) \cong (3,1) \otimes \mathrm{sgn}$ and $H_Q = (3,1)$ are distinct S_4-modules differing by a parity sign, but in three-dimensional geometry form an axial–polar vector duality via the Hodge star.

18.3 Boundaries That Must Be Stated

1. **Division of labor of Lovelock**: it only answers "what must the field equations look like," not "why does geometry have dynamics."
2. **Local K_4 fiber vs. macroscopic grid are two structures.** $\dim H_Q = m-1 = 3$ holds only for the *single-point* local K_4 fiber; on the macroscopic N-vertex grid, $\dim H_Q = N - 1 \neq 3$. The construction bridge "how local K_4 fibers/cells tile a macroscopic 3D manifold grid" is missing; it is an open item.
3. $L = 4$ **uses Gate 1 only.** The peak condition locks $L = 4$; this $k = 2$ partition does not pass the contextuality gate or the density gate, so it serves only as dimension readout, not as a quantumness criterion.

---

19 Open Problem

Continuum limit (Γ-convergence) is unproven: no compactness, no Γ-liminf; only on a 1D chain with a single profile. This is analytic (compactness / variational limit) and requires independent work in homogenization theory.

---

20 Conclusion

$$
\begin{aligned}
&\text{substrate (zero-sum constraint)}\\
&\quad\Longrightarrow \dim H_Q = m-1;\ \operatorname{im} B = H_Q;\ \text{discrete Dirichlet energy;}\\
&\quad\Longrightarrow (\text{continuum + metric-volume consistency}) \text{ spatial metric } h;\\
&\quad\Longrightarrow (\text{reversible/irreversible splitting}) \text{ signature } (1, m-1);\\
&\quad\Longrightarrow (\text{locality} \wedge \text{second-order} \wedge \text{conserved source})\\
&\quad\Longrightarrow G_{ab} + \Lambda g_{ab} = 8\pi G\, T_{ab} \quad (\text{in } D = 4).
\end{aligned}
\tag{10.1}
$$

Shape is unconditional; entry is conditional: $(L)\wedge(O)\wedge(C') \Rightarrow$ Einstein form is unconditional (Lovelock); dimension, metric, and source are three entries, all non-consequences of the postulates.

Current progress is **conditional recovery**, not **unconditional derivation**. To reach *derivation*, the same ledgers are missing: the premise of dimension, the origin of the metric, and the price tag of the source.

References

[1] R. Penrose, *The Road to Reality*, Vintage Books, 2005.

[2] C. J. Isham, "Structural issues in quantum gravity," arXiv:gr-qc/9510034, 1995.

[3] D. Lovelock, "The Einstein tensor and its derivatives," *J. Math. Phys.* **12**, 498 (1971).

[4] W. Feller, *An Introduction to Probability Theory and Its Applications*, Vol. 1, 3rd ed., Wiley, 1968.

[5] A. Schrijver, *Theory of Linear and Integer Programming*, Wiley, 1986.

[6] M. P. do Carmo, *Riemannian Geometry*, Birkhäuser, 1992.

[7] D. G. Cioranescu, P. Donato, *An Introduction to Homogenization*, Oxford University Press, 1999.

[8] J. D. Murray, *Mathematical Biology*, Vol. II: Spatial Models and Biomedical Applications, 3rd ed., Springer, 2003.

[9] H. Weyl, *Classical Groups: Their Invariants and Representations*, Princeton University Press, 1939.

[10] M. J. Duff, L. B. Okun, G. Veneziano, "Trialogue on the number of fundamental constants," *JHEP* **0203**, 023 (2002), arXiv:physics/0110060.